Efficient Convolutional Neural Architecture Search for LiDAR DSM Classification

Aili Wang, *Member, IEEE*, Dong Xue, Haibin Wu, and Yanfeng Gu, *Senior Member, IEEE*

***Abstract*— Light detection and ranging (LiDAR) data provide rich elevation information, so it plays an irreplaceable role in ground object classification. Recently, convolutional neural networks (CNNs) have shown excellent performance in LiDAR digital surface models (DSMs) classification. However, the architecture of CNN model relies heavily on manual design, so it has great limitations. In addition, different sensors capture LiDAR datasets with different properties, so the model should be designed to suit for different datasets, which further increases the workload of architecture design. Therefore, this article proposes a method of automatic design of LiDAR DSM classification model. First, attention mechanism is introduced into search space to improve the feature extraction capability of the model. Then, a gradient-based search strategy is used to obtain the optimal architecture from this search space. Second, a learning rate adjustment strategy is proposed to reduce the time spent in the search stage and evaluation stage to improve the classification accuracy of the model. Finally, a regularization scheme is introduced to enhance the robustness of the model and avoid overfitting. Experimental results on three public LiDAR datasets (Bayview Park, Recology, and Houston) obtained from different sensors show that the proposed neural architecture search method achieves the impressive classification performance compared to several state-of-the-art classification methods and improves the classification accuracy under the condition of limited training samples.**

***Index Terms*— Attention mechanism, convolutional neural networks (CNNs), digital surface models (DSMs), light detection and ranging (LiDAR), neural architecture search (NAS).**

I. Introduction

LIGHT detection and ranging (LiDAR) is a remote sensing method that uses light in the form of a pulsed laser to measure ranges (variable distances) to the Earth.

These light pulses combined with other data recorded by the airborne system, which generates precise 3-D information about the shape of the Earth and its surface characteristics [1]. LiDAR systems allow scientists and mapping professionals to examine both natural and man-made environments with accuracy, precision, and flexibility. The scientists are using LiDAR to produce more accurate shoreline maps, make digital elevation models for use in geographic information systems, and assist in emergency response operations, which has been widely used in many fields, such as coastal zone monitoring [2], canopy coverage estimation [3], urban planning [4], and Doppler measurement [5]. Most of the existing LiDAR data classification methods are based on point cloud data [6]–[8], but this article mainly studies the classification of digital surface model (DSM). DSM data can be obtained from point cloud data through a series of processing such as denoising and rasterization, which records the elevation information of surface features to carry out the terrain classification task [9].

The LiDAR DSM classification task is usually completed based on pixel-level methods. For example, some traditional methods include support vector machine (SVM) [10], decision tree [7], random forest (RF) [11], and other classification methods. Recently, with the continuous development of deep learning (DL), its advantages in image processing have been gradually highlighted, especially convolutional neural network (CNN). CNN is an end-to-end global feature extraction framework with strong robust feature extraction ability, which makes it superior in image classification tasks. Therefore, various CNN-based models have been applied to LiDAR DSM classification tasks, which gradually becomes the mainstream methods [12], [13]. For example, He *et al.* [14] combined CNN with spatial transformation network (STN) to determine the optimal input of LiDAR data and further used morphological profiles (MPs) to improve the classification accuracy. Wang *et al.* [15] proposed to combine octave convolution (Octconv) with lightweight SqueezeNet to form a dual neural network structure, which greatly reduced the number of network model parameters while not decreasing the classification accuracy. Ghamisi *et al.* [16] combined CNN and extension profiles (EPs) to extract features, which improved the classification accuracy of LiDAR data. Wu *et al.* [17] combined Octconv with capsule network to extract the high- and low-frequency features of spatial information from LiDAR data and further improved the classification accuracy.

Most of the above LiDAR DSM classification methods trained a new CNN model from scratch to complete the classification purpose. However, the CNN model usually has a huge amount of parameters, so it needs a large amount of manual labeled data to achieve a better classification effect [18]. However, in the field of remote sensing image processing, the number of labeled samples is limited and the cost of manual labeling is very high. It becomes more difficult to use specific remote sensing data to train the deep CNN model. Therefore, in order to fully train the network to

Manuscript received November 12, 2021; revised March 15, 2022; accepted April 28, 2022. Date of publication May 2, 2022; date of current version May 13, 2022. This work was supported by the Natural Science Foundation of China under Grant 61671190. *(Corresponding author: Haibin Wu.)*

Aili Wang, Dong Xue, and Haibin Wu are with the Heilongjiang Province Key Laboratory of Laser Spectroscopy Technology and Application, College of Measurement and Control Technology and Communication Engineering, Harbin University of Science and Technology, Harbin 150080, China (e-mail: aili925@hrbust.edu.cn; 1920600040@stu.hrbust.edu.cn; woo@hrbust.edu.cn).

Yanfeng Gu is with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: guyf@hit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3171520

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

improve the classification accuracy, there exist two mainstream methods. One method is based on transfer learning, which fully trains the network model on the dataset with a large number of labeled samples (such as ImageNet) and then transfers it to the remote sensing dataset with small sample size to complete effective classification. These networks include the generalization models of CNN such as GoogleNet [19], ResNet [20], DenseNet [21], and VGGNet [22], which also effectively alleviates the overfitting phenomenon. Another method is to use data enhancement processing or some special networks to generate false data to increase the amount of training sample data, so as to fully train the classification network model and improve the learning ability [23], [24]. For example, Wang *et al.* [24] combined the conditional generative adversarial network (CGAN) with the residual cell to increase the depth of the generator to obtain realistic fake samples. Then, DropBlock is used to randomly discard the feature region to greatly improve the accuracy of the discriminator in the classification of LiDAR data [24]. Wang *et al.* [25] proposed an end-to-end few-shot synthetic aperture radar (SAR) automatic target recognition method. This method achieves good recognition performance when there are few target training samples to be recognized. Wang *et al.* [26] also proposed a hybrid inference method, which can effectively utilize the supervised information of labeled samples and the manifold structures of the unlabeled samples. In addition, an enhanced hybrid loss function is proposed to constrain the samples so as to obtain better separability between classes in the embedding space. Experimental results demonstrate the effectiveness and robustness of the proposed method in few-shot SAR automatic target recognition tasks [26]. Yang *et al.* [27] proposed a mixed loss graph attention network. This method augments the SAR images, maps them to low-dimensional embedding space by the embedding network, and outputs the labels by attention-enhanced graph attention network (GAT). The mixed loss is also used to gain strong similarity for intraclass. Compared with other methods, this method has faster convergence speed, higher classification accuracy, and stronger robustness to large depression angle variation [27].

However, different remote sensing datasets are obtained by sensors with different resolutions, and each dataset should correspond to different network models due to its own characteristics. However, the above network is not universal to different datasets, which are little time-consuming and laborious manually designed by experts through experiments and experience. Based on these circumstances, researchers aim to find a method to automatic design network architecture to meet the special application needs, such as image classification and image semantic segmentation. Therefore, neural architecture search (NAS) technology came into being and attracted more and more attention [28].

NAS mainly includes three parts: search space, search strategy, and performance estimation strategy. The search space can be represented as a collection of all basic operations. Defining an appropriate search space can reduce its size and generate a new network structure. There are two main types of search space. The first is to construct the global search space of the whole neural network structure, including chain-structured search space [29] and multibranch structure search space [30]. The second is to stack specific repetitive structures to construct the cell-based search space of neural network structure [31]. The search strategy can automatically find the best performance of the neural network design, which includes random search [32], [33], Bayesian optimization [34], [35], evolutionary algorithm (EA) [36], [37], reinforcement learning (RL) [38], [39], and gradient-based methods [40], [41].

In the NAS domain, the mainly used strategies are EA, RL, and gradient-based scheme. EA comes from the natural evolution process, and the neural network structure is expressed as genes, which searched the network structure through a variety of mutations, cross, and select operations. At present, EA-based methods only optimize the network structure and then use the stochastic gradient descent (SGD) method to optimize the weight through backpropagation. For example, Real *et al.* [42] generated younger genotypes by introducing age attribute and designed AmoebaNet based on the EA model, which achieved good classification results on several public datasets. However, EA tends to evolve an architecture group to ensure the diversity of random uncontrollable mutations, but this mutation makes the evolution speed of EA slower and reduces the efficiency.

The RL-based method optimizes the reward function by using agents to generate different candidate architectures. Therefore, it is very important to design an appropriate reward function to guide the optimization process. Different RL-based methods use different agent strategies and optimize strategies. For example, Zoph and Le [43] used recurrent neural network (RNN) agents to control operations between cells and find candidate architectures. The evolutionary error of the architecture is used as a reward to generate another better architecture. Cai *et al.* [44] adopted a Net2Net transformation strategy, which modifies the existing architecture through transformation operation, so there is no requirement to search and train the network from scratch. Although the RL-based method can build a stable architecture to evolve the network, it needs a lot of attempts to obtain the correct reward to update the architecture. Therefore, the RL-based method has too much computation in the training process.

Because the search space of the above two strategies is huge and discrete, they are very expensive and time-consuming in calculation. The gradient-based method makes the search space continuous and improves the search efficiency by converting the operations into the weighted sum of all candidate operations. The most typical strategy is the differentiable architecture search (DARTS) [40], which searches for the optimal architecture in the continuous and differentiable search space and then determines the final cell architecture through the gradient descent method.

The performance estimation strategy is used to evaluate the performance of the completed search network. Its core is weight parameter optimization, including individual optimization and weight sharing optimization. So far, the NAS methods are used in the processing of natural images, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

only a few are used in the classification task of LiDAR DSM data [45].

At present, most classification methods of remote sensing images are CNN-based models. Because CNN has the ability to automatically extract the features of data, which has achieved many impressive results in the classification task of remote sensing, and the deeper the architecture, the stronger the ability to extract a large number of spatial and channel features from remote sensing data. However, CNN also extracted many features which are less important. Therefore, the proposal of attention mechanism helps CNN solve this problem, which can give up focusing on useless features and focus on more important features, so as to strengthen the CNN's ability of feature extraction [46]–[51]. The attention mechanism comes from the attention thinking mode of human brain. The brain's ability to process information is usually limited. When people observe an image, they do not look at every pixel of the whole image at once. Most of them focus on specific parts of the image according to their needs. Moreover, humans will learn from the previously observed images where they should focus their attention in the future. Therefore, it is necessary to use limited attention resources to selectively extract high-value information and suppress the interference of other irrelevant information.

Therefore, many studies have combined the attention mechanism with CNN to improve the performance of CNN model. Wang *et al.* [46] proposed the residual attention network, which is composed of multiple attention modules stacked. Each module has the trunk branch and the mask branch, where the former extracts features and the latter constructs an attention mask and multiplies with the output of the former. Fu *et al.* [47] proposed dual attention networks (DANets) to adaptively integrate the correlation between local features and global features. DANet includes two types of attention modules, which establish semantic related models in spatial dimension and channel dimension. In recent years, attention mechanism has been widely used in the remote sensing data classification applications.

In this article, inspired by the NAS method and the attention mechanism, we study a new paradigm to complete the classification of LiDAR DSM data by automatically designing joint attention mechanism and CNN architecture.

The main contributions of the proposed method for LiDAR data classification are described as follows.

1) A method of using NAS to design network architecture to complete the LiDAR data classification task is studied in this article. A new search space is proposed to help the network pay attention to useful features and abandon useless features by introducing the attention module. Then, the DARTS strategy is used to find the optimal architecture in this continuous space.
2) In order to improve the efficiency of architecture search, a learning rate adjustment strategy called cosine power annealing CPA) is introduced to improve the reasoning speed of the model and the accuracy of LiDAR data classification and further reduce the required computational cost.
3) The effectiveness of this classification method is proven on three challenging public LiDAR datasets. A large number of experimental results on Bayview Park, Recology, and Houston datasets show that compared with other representative methods, the neural network architecture searched by our proposed method achieves the impressive classification results, and at the same time, the search efficiency is much higher.

The remaining part of this article is organized as follows. Section II introduces the principle of DARTS. Section III introduces our proposed method in detail. Section IV describes the LiDAR datasets and experimental results. Section V analyzes the optimal architecture and ablation study. Section VI discusses the conclusions and future work.

II. Principles of DARTS

The LiDAR data classification framework proposed in this article is described, as shown in Fig. 1, which introduces an attention mechanism into the search space to strengthen the feature extraction ability in the search and evaluation stage. The LiDAR data to be classified are used as input to the model of automatic architecture search CNN, which contains multiple normal cells and reduction cells. The architecture of each cell is obtained through the DARTS strategy. The operations between nodes include convolution operation and triplet attention (TA) mechanism. In addition, an effective learning rate adjustment strategy is proposed to improve the performance of the model, and a regularization scheme is used to alleviate overfitting phenomenon and generate a more robust model. In this section, we introduce the principle of DARTS method and analyze its limitations.

DARTS is a gradient-based search strategy and has shown its advantages in many image processing tasks. The DARTS strategy shows its great application potential by reducing the search cost from thousands of GPU days to several GPU days. At the same time, the strategy also opens up a new path for LiDAR data classification. The meanings of the mathematical symbols used in this article are shown in Table I.

DARTS forms the final architecture by searching a cell and stacking it continuously. The cell mainly includes two types: normal cell and reduction cell. In the normal cell, the step size of each operation between the two nodes is 1, resulting in the size of the output that is the same as the input image. In the reduction cell, the step size of each operation between the two nodes is 2 to reduce the height and width of the feature graph. The cell is a directed acyclic graph (DAG), which is composed of N ordered intermediate nodes with a potential representation $x^{(m)}$, where m represents the order in DAG. The DAG contains multiple directed edges to connect nodes, and there are candidate operations $o^{(m,n)}$ associated with each directed edge, which transmit information from $x^{(m)}$ to $x^{(n)}$. Therefore, the output of $x^{(n)}$ is calculated according to all nodes before it in the same cell, as shown in the following formula:

$$x^{(n)} = \sum_{m<n} o^{(m,n)}(x^{(m)}). \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

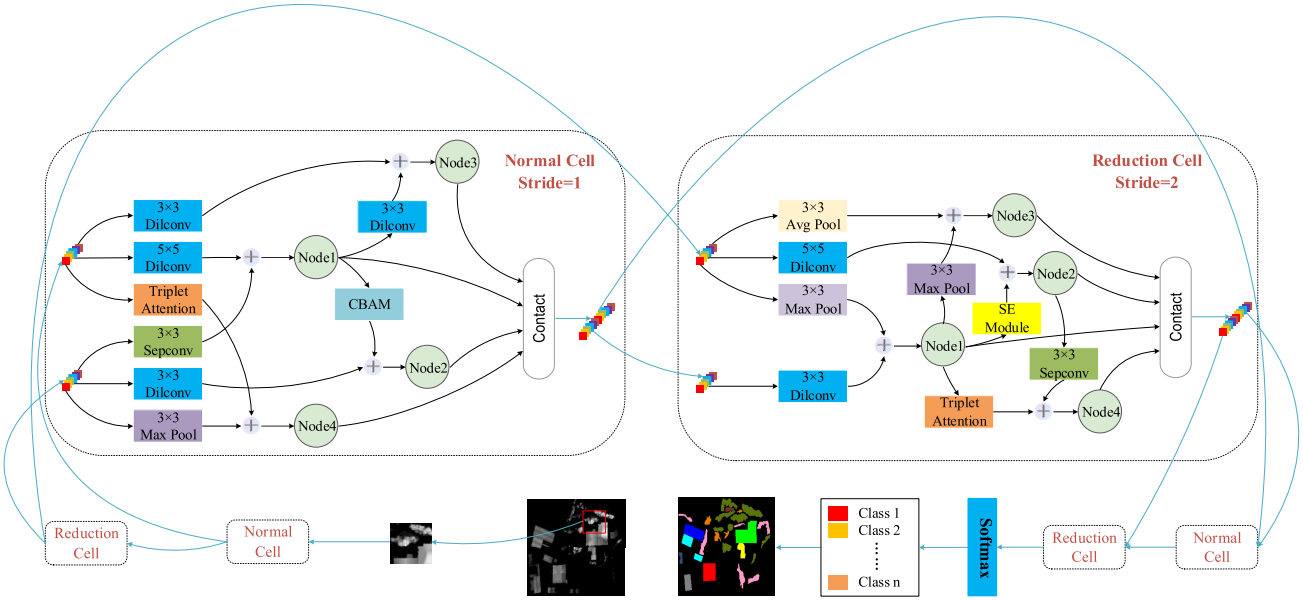


Fig. 1. Overall framework of the proposed LiDAR data classification model.

TABLE I
MATHEMATICAL SYMBOLS USED IN THIS ARTICLE

Symbol	Meaning
C	Cell
N	The number of nodes in a cell
$x^{(m)}$, $x^{(n)}$ $(m,n=0,1,\mathrm{L}\ ,N-1)$	A node in a cell
$\mathrm{O}=\{o_1^{(m,n)},o_2^{(m,n)},\mathrm{L}\ ,o_M^{(m,n)}\}$	Original candidate operation search space
$\mathrm{A}=\{a_1^{(m,n)},a_2^{(m,n)},\mathrm{L}\ ,a_M^{(m,n)}\}$	The attention mechanism searches space
$\mathrm{S}=\{\mathrm{A},\mathrm{O}\}$	New candidate operation search space
$p_o^{(m,n)}$	The operation coefficient of the original candidate operation
$\alpha_o^{(m,n)}$	Architecture parameters
$c_a^{(m,n)}$	The operation coefficient of the attentional mechanism
$\beta_a^{(m,n)}$	Attention mechanism parameter
$f_s^{(m,n)}$	Operation coefficient of mixed operation
$\gamma_s^{(m,n)}=\{\alpha_o^{(m,n)},\beta_a^{(m,n)}\}$	Mixed operation parameter
w	The weight parameters
$\mathrm{L}_{\mathrm{train}}$	Training loss
$\mathrm{L}_{\mathrm{val}}$	Validation loss

Let $O=\{o_1^{(m,n)},o_2^{(m,n)},\ldots,o_M^{(m,n)}\}$ be a set of candidate operation sets on the directed edge (e.g., convolution and max pooling). In the DARTS, the operations of each directed edge are parameterized by the architecture parameter $\alpha^{(m,n)}$. Therefore, in order to make the search space continuous, the discrete operations in the candidate operation set o are transformed into the set of Softmax algorithms for all possible operations by $\alpha^{(m,n)}$. In addition, the output of a set of candidate operation sets is the weighted sum of the output of each operation

$$\bar{o}^{(m,n)}(x)=\sum_{o\in O}\frac{\exp\left(\alpha_o^{(m,n)}\right)}{\sum_{o'\in O}\exp\left(\alpha_{o'}^{(m,n)}\right)}o(x). \tag{2}$$

Also, the sum of all factors in the operation set is 1. In addition, the weights of all hybrid operations between two nodes are obtained by vector $\alpha^{(m,n)}$ parameterization with dimension $|O|$. After each search, select the final operation between the two nodes according to $o^{(m,n)}=\arg\max_{o\in O}\alpha_o^{(m,n)}$, so as to obtain the structure of normal cell and reduced cell, and determine the final network architecture.

After relaxation, the goal of DARTS is to jointly learn the architecture parameters $\alpha=\{\alpha^{(m,n)}\}$ and network weights w in all hybrid operations. In the search process, L_{train} and L_{val} represent the training and verification losses, respectively. The ultimate goal of architecture search is to find the best architecture α^* to minimize the verification loss $L_{\mathrm{val}}(w^*,\alpha^*)$, in which the network weight w is obtained by minimizing the training loss L_{train}. This can be thought of as a bilevel optimization problem, which α is high-level variable and w is low-level variable

$$\min_{\alpha} L_{\mathrm{val}}\left(w^*(\alpha),\alpha\right) \tag{3}$$

$$\text{s.t. } w^*(\alpha)=\arg\min_{w} L_{\mathrm{train}}(w,\alpha). \tag{4}$$

Although DARTS is very efficient, it still has some limitations. First, almost all of its search space is convolution operation, and the traditional convolution operation will contain some useless features when it extracts features, resulting in that the waste features will affect the performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

of the final architecture. Second, DARTS aggregates multiple features generated by all candidate operations and updates all α and w in one step, which will lead to high computing cost and a lot of computing time. Therefore, we propose a new and efficient architecture search framework in Section III.

III. Ours Framework for LiDAR DSM Classification

The DARTS strategy shows its great application potential by reducing the search cost from thousands of GPU days to several GPU days. At the same time, the strategy also opens up a new path for LiDAR data classification.

A. DARTS With Attention Mechanism

Recent research on neural network architecture shows that the attention mechanism can improve the performance of neural network by discarding uninterested information, but the existing NAS methods pay little attention to it. Therefore, this article first proposes attention mechanism into convolutional NAS, called ACNAS model, which uses the attention module to form the final network architecture and uses cosine annealing to adjust the learning rate. Similar to most NAS methods, the whole network is stacked by repeated cells. Each cell C is a DAG composed of several directed edges and N nodes, consisting of two input nodes, one output node, and $N-3$ intermediate nodes. We define each node as $x^{(m)}(m = 0, 1, \ldots, N-1)$, which represents a feature graph. The information of the input node of the current cell is obtained from the output nodes of the previous two cells, and the output node is the depth cascade of all intermediate nodes in the cell. The directed edge in the cell represents the information flow from the intermediate $x^{(m)}$ to $x^{(n)}$, which is composed of candidate operations.

This proposed method searches for cells, including convolution operation and attention module, so we define a new attention space $A = \{a_1^{(m,n)}, a_2^{(m,n)}, \ldots, a_M^{(m,n)}\}$ for attention mechanism, which forms a new candidate operation set $S = \{A, O\}$ with the operation space O in the original DARTS. In the search phase, ACNAS selects the operation in O or the attention module in A from S to form the final operation between the two nodes. The search process of the cell is shown in Fig. 2.

For each intermediate node, select two nodes with the maximum value as its input and select an operation with the maximum weight value from the candidate operations as the final operation between the two nodes. The selection of final operation between nodes mainly includes three cases.

1) When the final operations between $x^{(m)}$ and $x^{(n)}$ are selected from O, we define an operation coefficient $p_o^{(m,n)}$ for each candidate operation, which is an all possible operations softmax mixture parameterized in O by the architecture parameters $\alpha_o^{(m,n)}$, as shown in (5). Therefore, the information flow from $x^{(m)}$ to $x^{(n)}$ is the weighted sum of all candidate operations, as shown in the following formula:

$$p_o^{(m,n)} = \frac{\exp\left(\alpha_o^{(m,n)}\right)}{\sum_{o' \in O} \exp\left(\alpha_{o'}^{(m,n)}\right)} \tag{5}$$

$$\bar{o}^{(m,n)}(x) = \sum_{o \in O} p_o^{(m,n)} o(x). \tag{6}$$

2) When the final operations between $x^{(m)}$ and $x^{(n)}$ are selected from A, we define an operation coefficient $c_a^{(m,n)}$ for each attention module, which is all possible attention module softmax mixtures parameterized in A by the attention module parameters $\beta_a^{(m,n)}$, as shown in (7). Therefore, the information flow from $x^{(m)}$ to $x^{(n)}$ is the weighted sum of all candidate attention modules, as shown in the following formula:

$$c_a^{(m,n)} = \frac{\exp\left(\beta_a^{(m,n)}\right)}{\sum_{a' \in A} \exp\left(\beta_{a'}^{(m,n)}\right)} \tag{7}$$

$$\bar{a}^{(m,n)}(x) = \sum_{a \in A} c_a^{(m,n)} a(x). \tag{8}$$

3) When the final operation between $x^{(m)}$ and $x^{(n)}$ is selected from both O and A, the final operation is a hybrid operation of the candidate operation and the candidate attention module. We define a hybrid operation coefficient $f_s^{(m,n)}$ for each hybrid operation, which is an all possible candidate operations and attention module softmax mixture parameterized in S by the hybrid operation parameters $\gamma_s^{(m,n)}$ (the combination of architecture parameters $\alpha_o^{(m,n)}$ and attention module parameters $\beta_a^{(m,n)}$), as shown in (9). Therefore, the information flow from $x^{(m)}$ to $x^{(n)}$ is the weighted sum of all hybrid operations, as shown in the following formula:

$$f_s^{(m,n)} = \frac{\exp\left(\gamma_s^{(m,n)}\right)}{\sum_{s' \in S} \exp\left(\gamma_{s'}^{(m,n)}\right)} \tag{9}$$

$$\bar{s}^{(m,n)}(x) = \sum_{s \in S} f_s^{(m,n)} s(x). \tag{10}$$

The output of cell C is obtained by connecting the output of each intermediate node, as shown in the following formula:

$$C = \sum_{m=1}^{N} \text{concat}(x^{(m)}). \tag{11}$$

After continuous relaxation, the architecture search is realized by optimizing the parametric network. The learning process of mixing operation parameter $\gamma = \{\alpha^{(m,n)}, \beta^{(m,n)}\}$ and weight parameter w can be expressed as a bilevel optimization problem as shown in (12) and (13). The learning process is shown in Fig. 3. The learned cells mainly include normal cells and reduced cells. Finally, through the first-order/second-order approximation, we can use the gradient-based method to solve this problem

$$\min_{\gamma} L_{\text{val}}\left(w^*(\gamma), \gamma\right) \tag{12}$$

$$\text{s.t. } w^*(\gamma) = \arg\min_{w} L_{\text{train}}(w, \gamma) \tag{13}$$

where γ is a high-level variable and w is a low-level variable.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

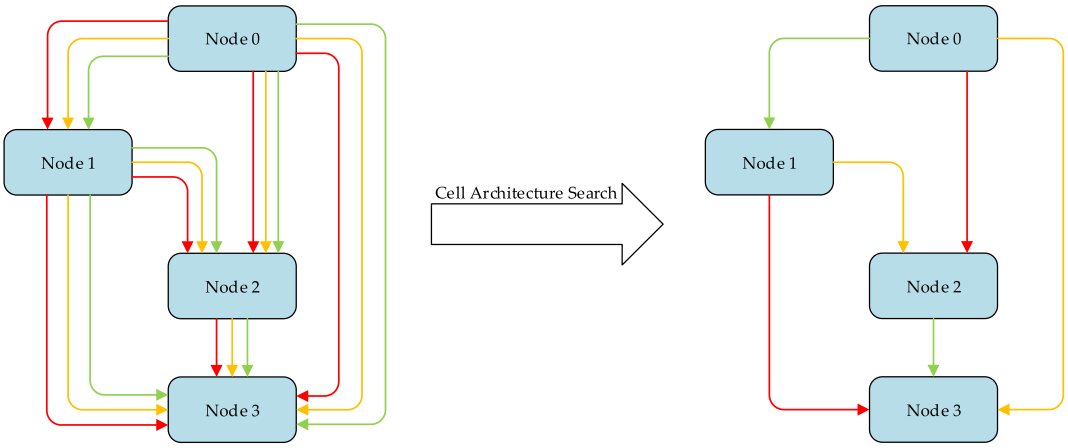


Fig. 2. Cell structure obtained based on the DARTS strategy. The cell contains four nodes, and the final operation between nodes is searched from the candidate operation space O and attention space A.

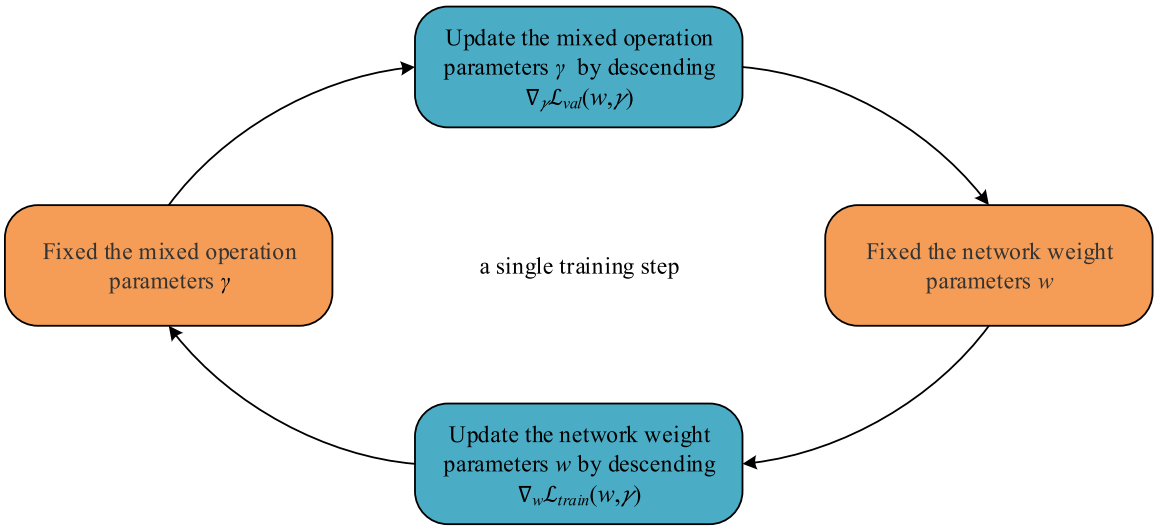


Fig. 3. Optimization of hybrid operation parameters γ and network weight parameters w based on gradient.

B. Learning Rate Adjustment Strategy

As discussed in Section III-A, DARTS consumes a lot of time and expensive computing costs in the search and evaluation phase. Therefore, we proposed a new learning rate adjustment strategy to reduce the time and cost, so as to help us find the optimal architecture suitable for each LiDAR data classification.

Most NAS methods use a cosine annealing strategy to adjust the learning rate. This strategy can reduce the learning rate through cosine function, so as to help the network approach this minimum value as much as possible.

The cosine function can adjust the learning rate with time, and its formula is given as follows:

$$\eta_t = \eta_{\min}^i + \frac{1}{2}\left(\eta_{\max}^i - \eta_{\min}^i\right)\left(1 + \cos\left(\pi \frac{T_{\text{cur}}}{T_i}\right)\right) \tag{14}$$

where η_t is the learning rate of the current number of epochs; $\eta_{\min}^i$ and $\eta_{\max}^i$ represent the minimum learning rate and the maximum learning rate, respectively; T_i is the total number of epochs; T_{cur} is the current number of epochs; and i is the index into a list of parameters for warm restarts in which $\eta_{\max}^i$ typically decays. In the experiment, it can be directly implemented without warm restarts, where $i = 0$.

Although cosine annealing performs well in DARTS, the decay rate of its learning rate is unbalanced, as shown in Fig. 4. The learning rate of cosine annealing decreases with a relatively slow decay rate at the beginning but then decreases with a relatively fast decay rate. This approach will lead to the imbalance of learning rate, resulting in DARTS spending a lot of time in the process of search and evaluation to improve the classification accuracy of the model finally searched. Therefore, we introduce a power curve parameter p into the original cosine function to reduce this imbalance, which is called CPA strategy and is defined as follows:

$$\eta_t = \eta_{\min}^i + \left(\eta_{\max}^i - \eta_{\min}^i\right)\frac{p^{\frac{1}{2}\left(1+\cos\left(\pi \frac{T_{\text{cur}}}{T_i}\right)\right)+1} - p}{p^2 - p} \tag{15}$$

where p should be greater than 1. This strategy adjusts the decay rate of the learning rate by introducing the normalized exponential term p so that the model has a higher learning rate in the first few epoch periods and adopts a shallow slope to reduce the learning rate in the last one-third of the epoch period. This parameter helps to reduce the time required by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

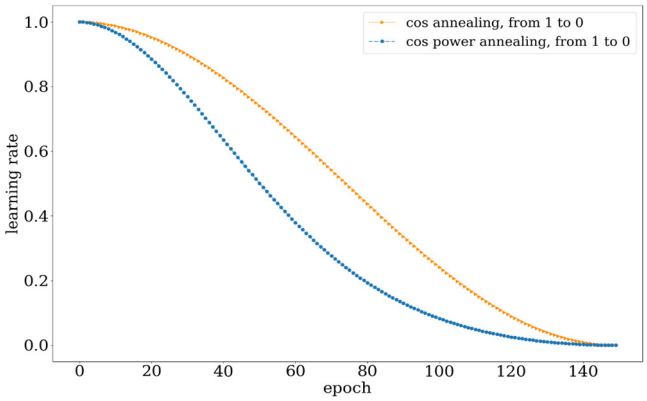


Fig. 4. Comparison of learning rate adjustment strategies of cosine annealing and CPA.

the model in the performance improvement stage, so as to further improve the classification accuracy of the model and reduce the computational cost. We combine the CPA strategy with the ACNAS model to form the ACNAS-CPA model.

C. Architecture Regularization Scheme

The network model searched by the NAS method usually contains a large number of learnable parameters, so it is prone to overfitting problems in the search and evaluation stage. In addition, the LiDAR dataset contains less training data, which will lead to insufficient training of network model and overfitting problem. Therefore, this article adds a label smoothing regularization scheme to the ACNAS-CPA model. Label smoothing is a regularization scheme commonly used in DL [52]. It can prevent the problem of overconfident prediction labels in the training of the model, so as to increase the generalization ability of the model.

In the training process, the traditional one-hot coded label will encourage the model to predict that the probability of the training sample as the target category approaches 1 and the probability of the nontarget category approaches 0. This coded label vector y_n can be written as

$$y_n = \begin{cases} 1, & n = q \\ 0, & n \neq q \end{cases} \tag{16}$$

where n represents the training sample's label. If the training samples contain N categories, then $n \in \{1, 2, \ldots, N\}$. q means the prediction of training samples. This method will make the model learn in the direction of infinite increase in the logit difference between correct and wrong labels, and too large logit difference will make the model lack adaptability and overconfident in its prediction. Therefore, when the training data are not enough to cover all cases, it will lead to overfitting and poor generalization ability of the network.

Label smoothing combines uniform distribution and replaces the traditional one-hot coded label vector y_n with the updated label vector $\hat{y}_n$. The smooth label distribution is computed as follows:

$$\hat{y}_n = \begin{cases} 1 - \grave{\mathrm{O}}, & n = q \\ \dfrac{\grave{\mathrm{O}}}{N-1}, & n \neq q \end{cases} \tag{17}$$

where Ò represents the smoothing factor. After label smoothing, the training sample is equivalent to adding noise to the sample, which avoids the model's overconfidence in the correct label and reduces the output difference between positive and negative samples. Label smoothing can also make the clusters between classes more compact, increase the distance between classes, and reduce the gap within classes, so as to avoid overfitting and improve the generalization ability of the model. Finally, we combine the label smoothing regularization scheme with ACNAS-CPA to form ACNAS-CPA-LS, which further improves the classification accuracy of LiDAR data.

IV. Experimental Results and Analysis

In this section, we summarize the experimental results and comparison analysis of LiDAR data classification task. First, we introduce the three datasets used in this article. Then, the classification performance of the proposed method is evaluated on three LiDAR datasets, and the comparison results with other six methods are reported. Finally, we give the classification maps, which shows the advantages of this model from the visual point of view.

The experiments are carried out in the windows operating system, classification approach is implemented using Python language and Pytorch library. The experimental equipment includes two 2.10-GHz CPUs and two NVIDIA RTX 2080Ti GPUs (11-GB memory). We used three indicators to quantitatively evaluate the final experimental results, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (K). OA represents the ratio of the correct number of pixels predicted by the model on the test set to the number of all pixels; AA refers to the average value of the ratio of the number of correctly predicted pixels of each category to the number of all pixels of each category; Kappa coefficient represents the difference between the classification results of the model and the completely random classification results, usually between 0 and 1. The larger kappa's value, the higher the classification accuracy.

A. Datasets Description

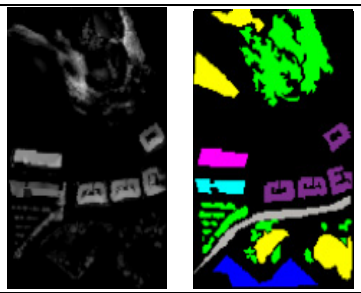
In this article, we use three public LiDAR datasets to evaluate the classification performance of the proposed model, namely, Bayview Park, Recology, and Houston datasets.

The Bayview Park consisted of 300 × 200 pixels with a spatial resolution of 1.8 m and the Recology consisted of 200 × 250 pixels with a spatial resolution of 1.8 m. They were all from the 2012 IEEE International Remote Sensing Image Convergence Competition and collected in the city of San Francisco, CA, USA. The Houston consisted of 349 × 1905 pixels with a spatial resolution of 2.5 m, which was collected by the National Science Foundation (NSF)-funded Center in the University of Houston Campus, Houston, TX, USA, on June 22, 2012.

Fig. 5 shows the DSM maps, the ground-truth maps, and land classes of interest of the three datasets.

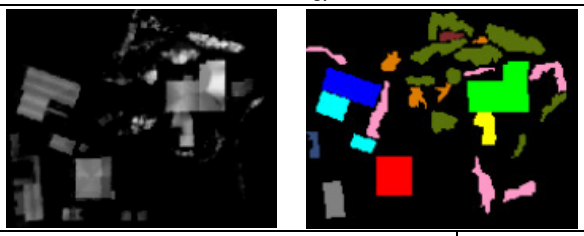
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

Bayview Park



Class			Samples
No.	Color	Name	Numbers
1		Building1	2282
2		Building2	719
3		Building3	995
4		Road	1566
5		Tress	7684
6		Soil	4283
7		Seawater	2008
Total			19537

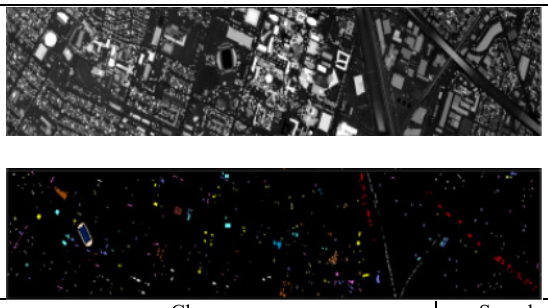
Recology



Class			Samples
No.	Color	Name	Numbers
1		Building1	1080
2		Building2	1136
3		Building3	1849
4		Building4	431
5		Building5	549
6		Building6	785
7		Building7	167
8		Trees	3321
9		Parking lot	1783
10		Soil	561
11		Grass	149
Total			11811

Fig. 5. (Top) Bayview Park dataset. (Middle) Recology dataset.

Houston



Class			Samples
No.	Color	Name	Numbers
1		Healthy grass	1251
2		Stressed grass	1254
3		Synthetic grass	697
4		Tree	1244
5		Soil	1242
6		Water	325
7		Residential	1268
8		Commercial	1244
9		Road	1252
10		Highway	1227
11		Railway	1235
12		Parking lot1	1233
13		Parking lot2	469
14		Tennis court	428
15		Running track	660
Total			15029

Fig. 5. *(Continued.)* (Bottom) Houston dataset.

B. Search Space Analysis

The search space of this experiment is mainly composed of two parts: candidate operation space O and attention mechanism space A. The initial candidate operation space $_O$ is the same as that in DARTS, which contains eight candidate operations, 3×3 separable convolution, 3×3 dilated separable convolution, 5×5 separable convolution, 5×5 dilated separable convolution, 3×3 max pooling, 3×3 average pooling, identity, and zero. Separate convolution can significantly reduce the number of network parameters, which divides the volume layer into two layers, namely, depthwise revolution and pointwise revolution, to improve the performance of the network [53]. Dilated convolution increases its receptive field by introducing the dilated rate into the convolution layer so that the feature map of the current layer can respond to a wider range of information [54]. The main function of the max pooling and average pooling is to refine the feature map and reduce its dimensions. The identity adds the feature map of the previous layer to the current layer by using shortcut connection, so as to help the network escape the local optimal solution in the backpropagation process [20]. Zero means that no operation is needed between two nodes.

The attention mechanism space A mainly includes four kinds of attention modules, including squeeze-and-excitation (SE) module, bottleneck attention module (BAM), convolutional block attention module (CBAM), and TA module. Fig. 6 shows the used attention module, where $\otimes$ represents elementwise multiplication and $\oplus$ represents elementwise summation.

SE can make the network focus on the channel. It recalibrates the response of the filter through two-step operations of squeezing and excitation, so as to enhance the learning of features in the convolution layer and improve the sensitivity of the network to information features. BAM combines channel attention with spatial attention, where channel attention is the same as that in the SE module. Spatial attention can make the network focus on important spatial positions. Its attention mask is the sum of the outputs of channel and spatial attention. CBAM is also a combination of channel attention and spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

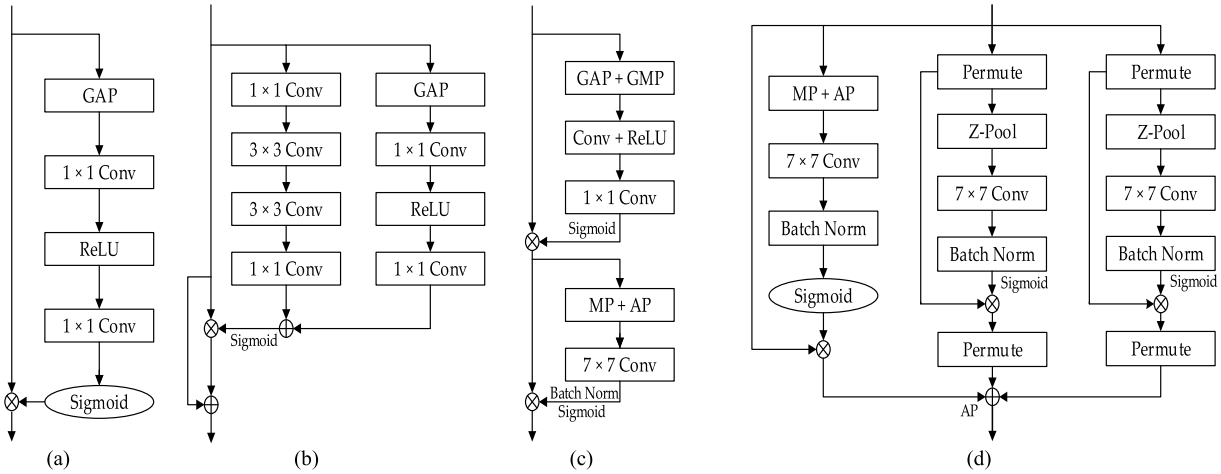


Fig. 6. Different attention modules. (a) SE. (b) BAM. (c) CBAM. (d) TA.

TABLE II
CLASSIFICATION RESULTS OF ALL METHODS ON THE BAYVIEW PARK DATASET

Methods	SVM	RF	CNN	ResNet	CNAS	ResCapNet	ACNAS-CPA-LS
Building1	81.96±2.56	94.28±2.47	93.49±1.53	98.04±1.58	99.95±0.05	99.47±0.83	100.00±0.00
Building2	85.51±2.37	97.69±1.28	93.83±1.82	99.51±1.49	100.00±0.00	99.86±0.23	100.00±0.00
Building3	91.49±4.19	99.21±0.94	92.69±2.17	99.62±2.26	100.00±0.00	100.00±0.00	100.00±0.00
Road	80.57±5.01	81.74±4.63	91.93±1.49	96.81±2.37	97.23±1.76	98.39±1.24	96.60±1.41
Tress	83.18±2.68	90.16±0.96	86.58±1.25	98.07±1.68	98.53±1.56	98.52±0.71	98.24±0.84
Soil	60.26±3.04	86.27±1.63	85.23±1.64	87.63±2.19	89.26±2.13	90.09±2.83	96.25±0.91
Seawater	84.63±1.76	82.25±2.15	90.61±1.97	90.76±2.38	90.97±2.47	93.49±2.18	92.53±2.04
OA(%)	76.18±1.92	89.87±1.32	90.25±2.38	94.69±1.18	95.53±1.04	96.27±0.73	97.32±0.57
AA(%)	81.09±1.36	90.23±1.57	90.62±1.59	95.78±1.54	96.56±1.12	97.12±1.25	97.66±1.31
K×100	68.59±2.15	86.23±1.02	88.76±2.43	93.49±1.47	94.37±1.29	95.19±0.87	96.87±0.69

TABLE III
CLASSIFICATION RESULTS OF ALL METHODS ON THE RECOLOGY DATASET

Methods	SVM	RF	CNN	ResNet	CNAS	ResCapNet	ACNAS-CPA-LS
Building1	72.16±1.21	90.82±3.08	98.25±1.07	98.51±1.37	98.42±1.17	98.39±1.42	99.12±0.75
Building2	65.04±1.39	95.19±4.05	95.26±1.29	98.02±1.68	98.68±1.31	99.74±0.26	100.00±0.00
Building3	93.09±1.17	94.03±0.83	94.17±0.96	98.18±1.84	96.83±1.64	98.49±1.13	99.40±0.54
Building4	89.97±2.36	96.38±1.48	97.15±1.24	95.42±1.72	98.96±1.53	99.61±1.24	99.33±1.69
Building5	85.51±1.13	96.22±2.32	96.53±1.91	98.77±1.08	99.14±1.04	99.03±0.98	99.78±0.85
Building6	69.25±0.97	93.51±1.28	94.71±0.94	96.19±2.17	97.29±1.39	98.52±1.57	99.56±2.93
Building7	87.51±1.85	96.39±2.36	96.83±1.77	93.24±2.58	97.61±1.68	98.73±1.28	99.49±1.67
Trees	86.87±1.26	95.19±1.61	95.41±1.28	97.18±1.29	98.07±1.43	95.41±1.83	98.72±2.53
Parking lot	64.24±1.49	76.61±0.84	76.59±1.79	90.33±2.23	90.64±1.56	91.27±1.71	92.82±3.16
Soil	81.21±3.24	72.48±0.57	72.36±1.26	89.21±2.11	93.13±2.37	95.69±2.21	94.99±3.14
Grass	97.94±2.09	98.24±2.07	96.47±1.34	92.54±2.49	97.05±2.28	95.87±2.44	98.47±1.62
OA(%)	78.15±1.03	91.31±0.81	92.49±1.25	95.83±0.97	96.36±1.14	96.84±0.92	98.08±0.76
AA(%)	81.16±1.32	91.37±1.48	92.16±1.27	95.24±2.04	96.89±1.52	97.34±1.07	98.33±1.41
K×100	73.88±1.25	89.12±1.09	90.93±1.61	95.14±1.06	95.81±1.39	96.57±0.66	97.51±0.58

attention. On the basis of global average pooling, it aggregates the spatial features of the input feature map by maximizing TA, including three parallel branches. The first branch is used to pooling. CBAM adds spatial attention after channel attention, connects their outputs, obtains the mask of spatial attention, and establishes spatial attention, and the other two

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION RESULTS OF ALL METHODS ON THE HOUSTON DATASET

Methods	SVM	RF	CNN	ResNet	CNAS	ResCapNet	ACNAS-CPA-LS
Healthy grass	80.16±3.84	86.37±2.57	88.68±3.14	91.94±2.07	93.17±1.97	95.09±1.84	96.16±2.25
Stressed grass	74.84±2.94	86.19±3.09	88.65±2.88	90.37±1.57	92.54±1.04	95.82±1.36	95.57±1.12
Synthetic grass	90.16±2.26	93.41±1.84	95.86±0.72	97.09±1.46	98.27±0.59	99.41±0.22	99.52±0.38
Tree	88.04±3.73	89.84±2.89	91.18±1.94	91.66±2.37	92.40±2.51	93.37±2.13	93.91±2.57
Soil	79.51±4.14	87.04±3.76	89.07±3.27	90.17±2.84	93.51±1.64	96.58±1.71	98.08±1.69
Water	90.04±1.65	95.71±1.31	97.31±0.71	97.84±0.42	98.69±0.87	99.01±0.32	98.52±0.47
Residential	82.64±3.21	85.92±3.39	88.09±3.08	90.79±2.01	92.66±2.14	95.36±1.34	95.51±1.29
Commercial	87.66±3.69	89.49±2.64	93.54±3.37	94.58±2.51	95.72±1.84	96.08±2.09	97.18±0.64
Road	78.97±2.24	84.72±2.13	87.36±2.76	88.95±1.57	90.44±2.19	91.29±1.97	93.52±1.53
Highway	91.24±2.61	96.94±2.38	98.07±1.15	98.24±0.94	98.89±0.87	99.87±0.13	100.00±0.00
Railway	94.79±1.19	96.51±1.33	96.01±0.48	96.79±0.97	97.38±0.82	98.85±0.94	99.92±0.07
Parking lot1	80.27±2.28	88.08±1.72	89.82±1.99	91.04±3.27	92.91±2.23	95.09±1.17	97.57±1.15
Parking lot2	84.11±2.34	92.79±1.19	98.24±1.44	98.61±1.31	98.94±0.82	99.13±0.49	99.47±0.84
Tennis court	95.62±0.91	100.00±0.00	99.53±0.27	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
Running track	92.13±3.93	96.08±1.82	93.49±2.16	95.91±1.72	97.42±2.44	98.07±1.82	99.61±0.29
OA(%)	84.92±0.94	89.14±0.31	91.51±0.59	93.72±0.35	94.97±0.22	96.29±0.19	97.12±0.21
AA(%)	86.01±0.72	91.27±0.58	92.99±0.26	94.27±0.64	95.53±0.43	96.87±0.24	97.64±0.31
K×100	84.35±0.59	88.25±0.69	91.04±0.38	92.99±0.52	94.83±0.67	96.19±0.37	96.85±0.28

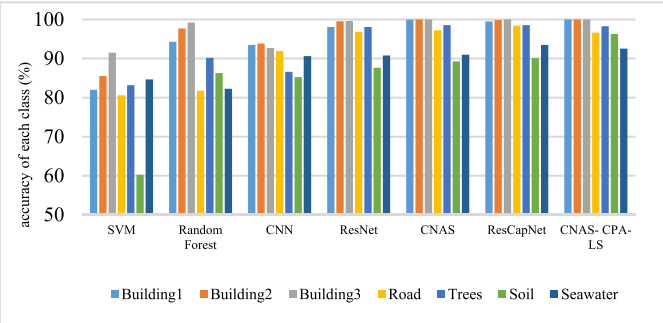


Fig. 7. Different methods' classification results for each class on the Bayview Park dataset.

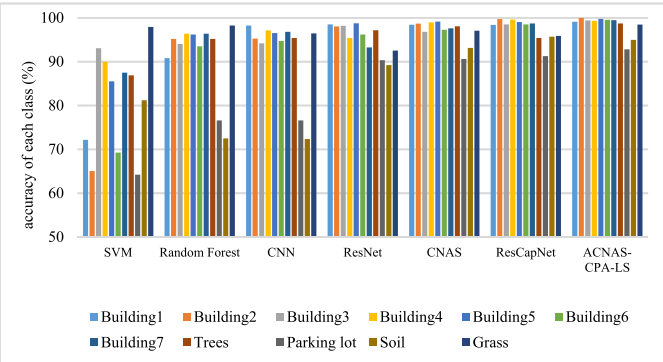


Fig. 8. Different methods' classification results for each class on the Recology dataset.

are used to capture the cross-dimensional interaction between channel dimension and spatial dimension. The final output is the average aggregation of the outputs of the three branches.

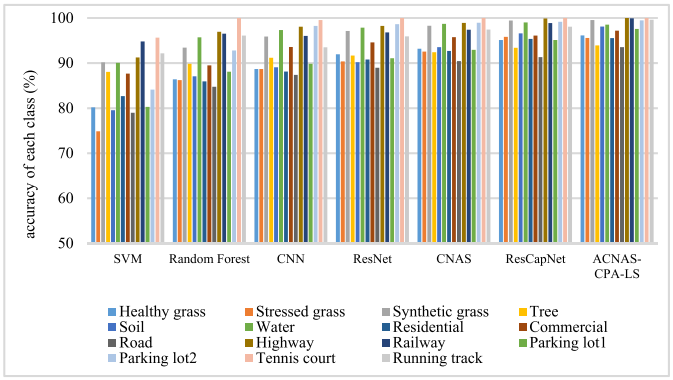


Fig. 9. Different methods' classification results for each class on the Houston dataset.

C. Implementation Details

First, LiDAR data are preprocessed and linearly mapped to $[-0.5, 0.5]$. Then, we adjust the size of the input sample of the model to a fixed size of 32×32 and randomly select 70 samples each class as the training set, 30 samples each class as the validation set, and the remaining samples as the test set. In order to obtain the optimal network model, we first use the search algorithm to learn the structure of cells on the training set and validation set of LiDAR datasets. When the optimal architecture is determined, the respective models are trained and evaluated from scratch.

In the search phase, we set the number of cells to 2 and the initial number of channels to 16. Each cell consists of seven nodes, including two input nodes, four intermediate nodes, and one output node. The search epoch of Bayview Park dataset, Recology dataset, and Houston dataset is set to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

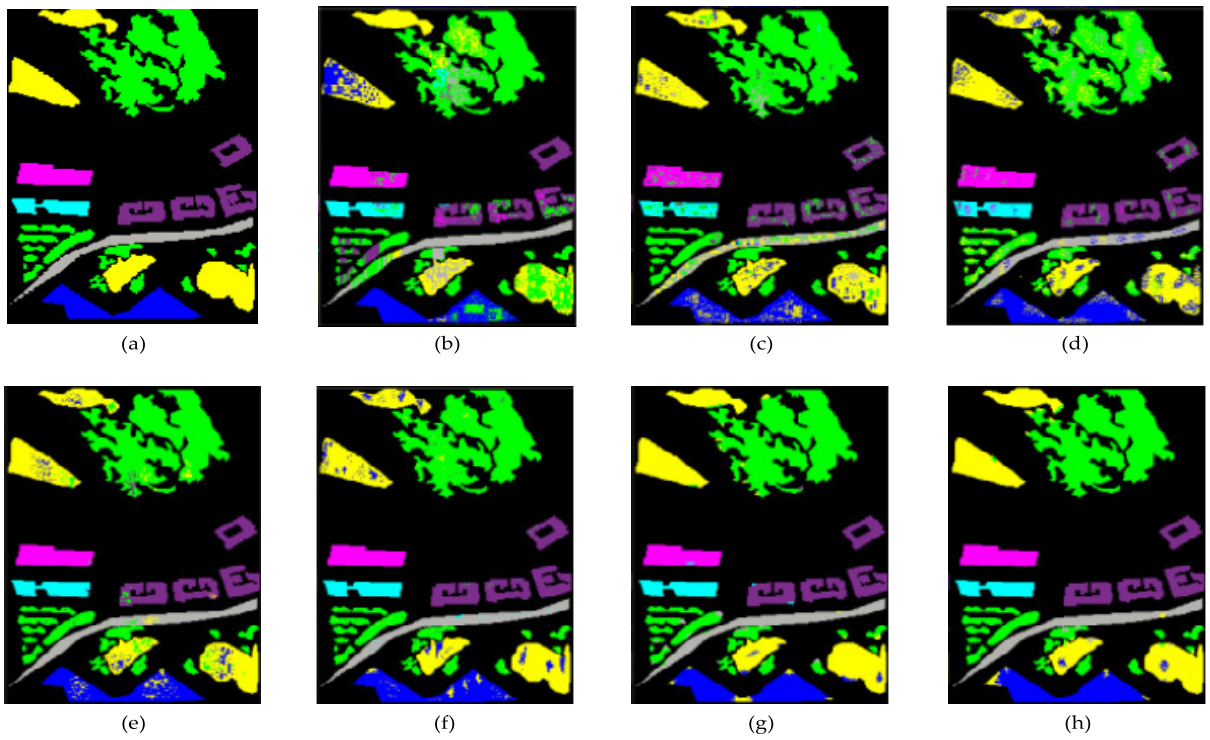


Fig. 10. Classification maps on the Bayview Park dataset. (a) Ground truth. (b) SVM. (c) RF. (d) CNN. (e) ResNet. (f) CNAS. (g) ResCapNet. (h) ACNAS-CPA-LS.

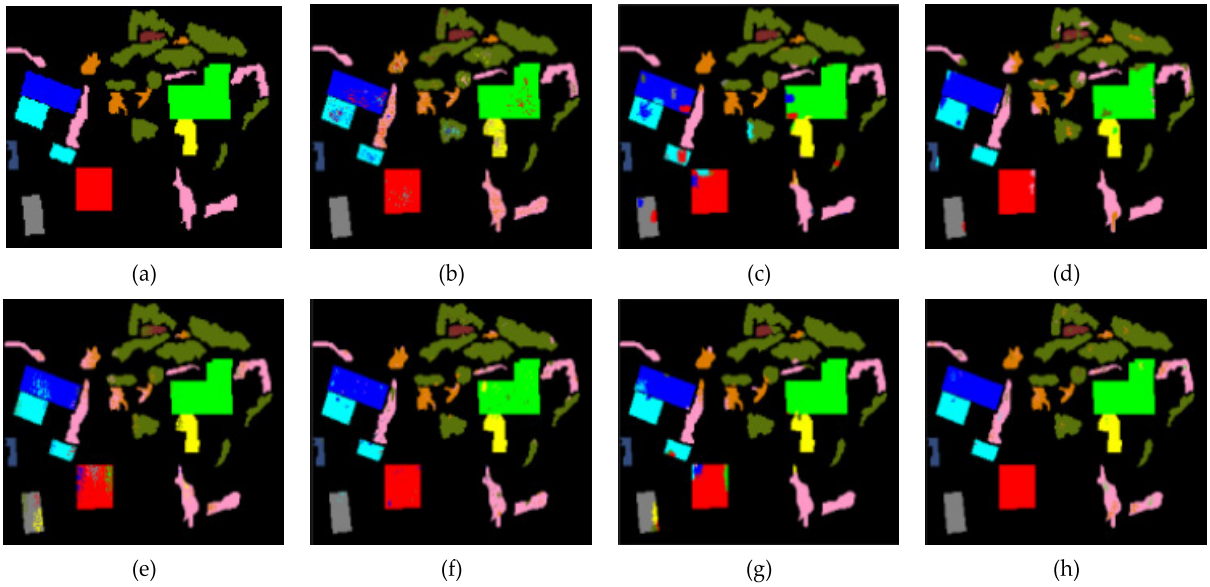


Fig. 11. Classification maps on the Recology dataset. (a) Ground truth. (b) SVM. (c) RF. (d) CNN. (e) ResNet. (f) CNAS. (g) ResCapNet. (h) ACNAS-CPA-LS.

150, 200, and 250, respectively. The batch size of each search phase is set to 32. We used the Adam optimizer to learn the mixed operation parameter γ, where learning rate = 0.0003, weight decay = 0.001, and momentum = (0.5, 0.999). The SGD is used to optimize the weight parameters of the model, where learning rate = 0.025, momentum = 0.9, and weight decay = 0.0003. For three datasets, the CPA strategy is used to reduce the learning rate, in which the maximum learning rate is set to 0.025, the minimum learning rate is set to 0.0001, and the power curve parameter p is set to 2.

In the evaluation stage, we stack the searched cells to form 4, the initial number of channels is set to 32, and each cell also includes seven nodes. Training the model from scratch, where the final network model, in which the number of evaluation epoch of Bayview Park, Recology, and Houston datasets is set to 200, 250, and 300, respectively. The batch size of each search phase is set to 64 to ensure the final convergence of the model. The CPA strategy is also used to reduce the learning rate. For the Bayview Park and Recology datasets, the maximum learning rate is set to 0.01 and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

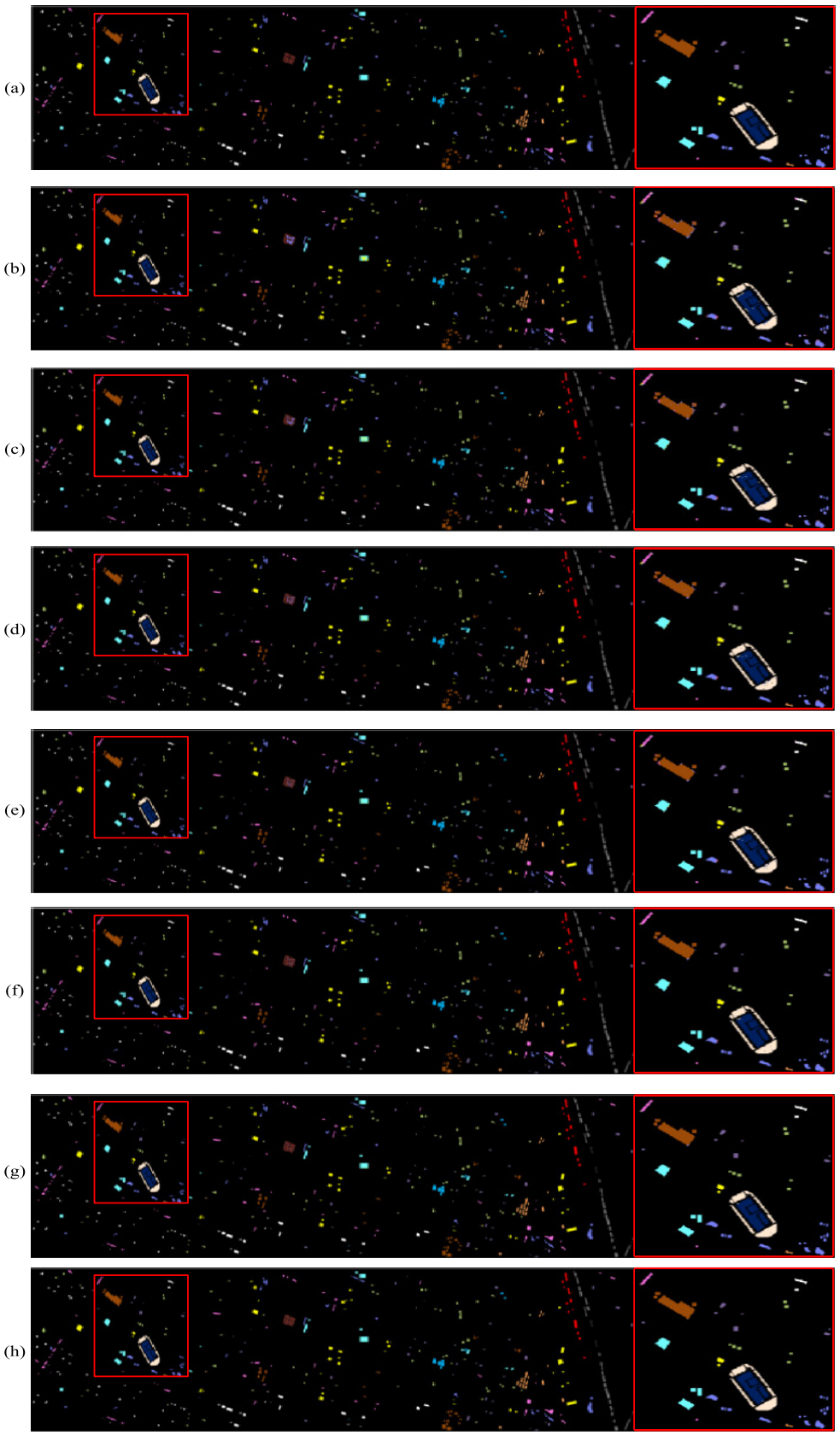


Fig. 12. Classification maps on the Houston dataset (took the most representative part). (a) Ground truth. (b) SVM. (c) RF. (d) CNN. (e) ResNet. (f) CNAS. (g) ResCapNet. (h) ACNAS-CPA-LS.

minimum learning rate is set to 0.001. For the Houston dataset, the maximum learning rate is set to 0.025 and the minimum learning rate is set to 0.001. Other configurations and super parameter settings are the same as those in the search phase.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

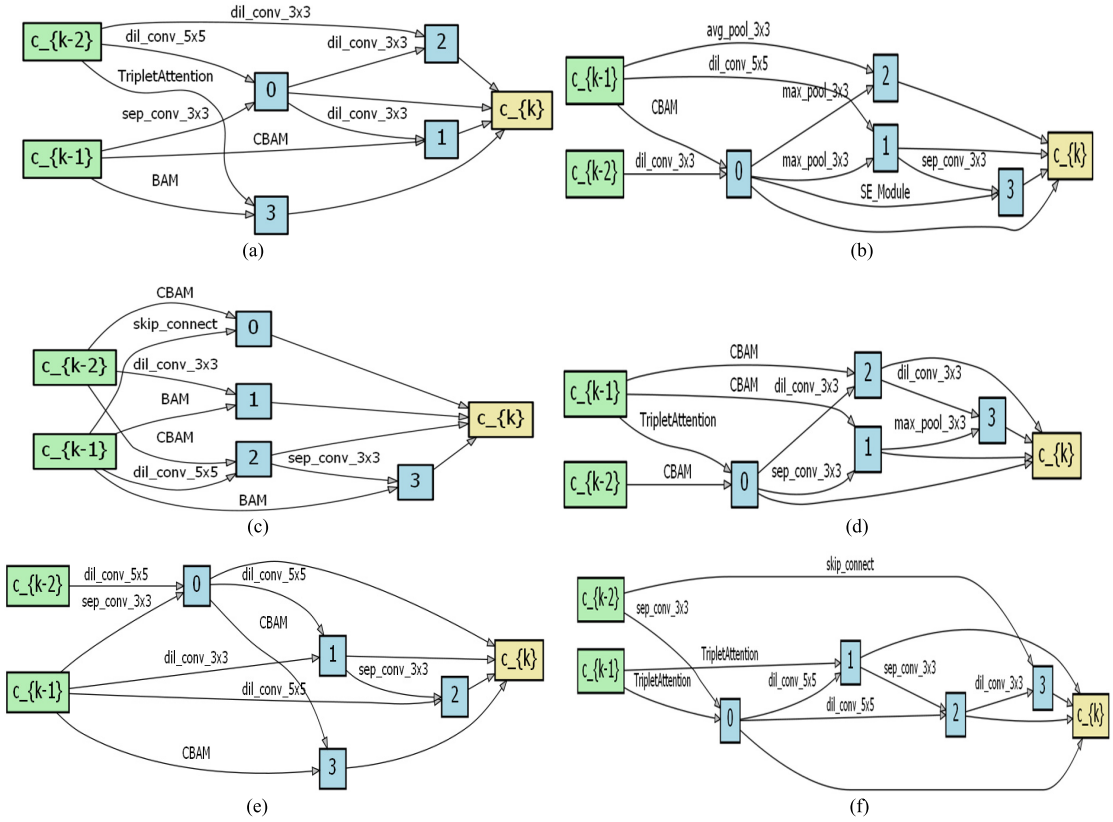


Fig. 13. Searched cell architectures of (Top) Bayview Park dataset, (Middle) Recology dataset, and (Bottom) Houston dataset. (a), (c), and (e) Searched normal cell. (b), (d), and (f) Searched reduction cell.

Finally, in order to ensure the robustness of the final model, the label smoothing regularization is used to avoid the overfitting phenomenon, in which the smoothing factor is set to 0.1.

D. Experimental Results

In order to evaluate the classification performance of our proposed method, we selected some methods to make a comprehensive comparison with ACNAS-CPA-LS, including two classical machine learning methods, SVM, and RF; and three manually designed models, CNN, ResNet, and ResCapNet. In addition, it also includes a method of automatically designing CNN model using NAS, which is called CNAS. The kernel function of the SVM was the radial basis function (RBF). The estimates of the RF for the two datasets were set to 30. For all manually designed CNN models, CNN, ResNet, and ResCapNet, we set the number of epochs to 300 and the learning rate of the two datasets to 0.001. For CNAS, it searches for the optimal architecture based on the DARTS strategy. Other configurations and super parameter settings are the same as ACNAS-CPA-LS.

Tables II–IV shows the detailed classification results of our proposed method and other comparison methods on the three datasets. Obviously, ACNAS-CPA-LS obtains better classification performance than other six comparison methods. It has the highest classification accuracy on the Bayview Park, Recology, and Houston datasets, in which OA has reached 97.32%, 98.08%, and 97.12%, respectively. For SVM and RF, their performance is limited by a few labeled samples, so the classification accuracy is the lowest. On the Bayview Park dataset, ACNAS-CPA-LS increased OA by 21.14% and 7.45% compared with SVM and RF, respectively. On the Recology dataset, ACNAS-CPA-LS increased the OA of SVM and RF by 19.93% and 6.77%, respectively. On the Houston dataset, ACNAS-CPA-LS increased the OA of SVM and RF by 12.20% and 7.98%, respectively. Although manually designed CNN and ResNet can use labeled samples, they do not fully learn all the features of the data due to the small number of training sets, so they cannot achieve high classification accuracy.

The CNN's OA on three datasets reached 90.25%, 94.69%, and 91.51%, and the ResNet's OA on three datasets reached 92.49%, 95.83%, and 93.72%. For the automatically designed CNAS model, it adopts different architectures for each dataset and obtains better classification performance. For ResCapNet, it is a combination of two models manually designed by experts, so it has stronger ability to extract features and higher classification accuracy. However, its classification effect is not as good as ACNAS-CPA-LS because its architecture is not necessarily suitable for every type of data. Also, our proposed method applies the DARTS strategy to the well-designed search space, which avoids the influence of overfitting phenomenon and unbalanced distribution of samples on classification, so it effectively improves the classification accuracy.

Figs. 7–9 show the classification results of each class by different methods. We can see that our method has almost reached the highest classification accuracy on each class for three LiDAR datasets. Especially for the categories with low height, its classification accuracy is improved, such as soil

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

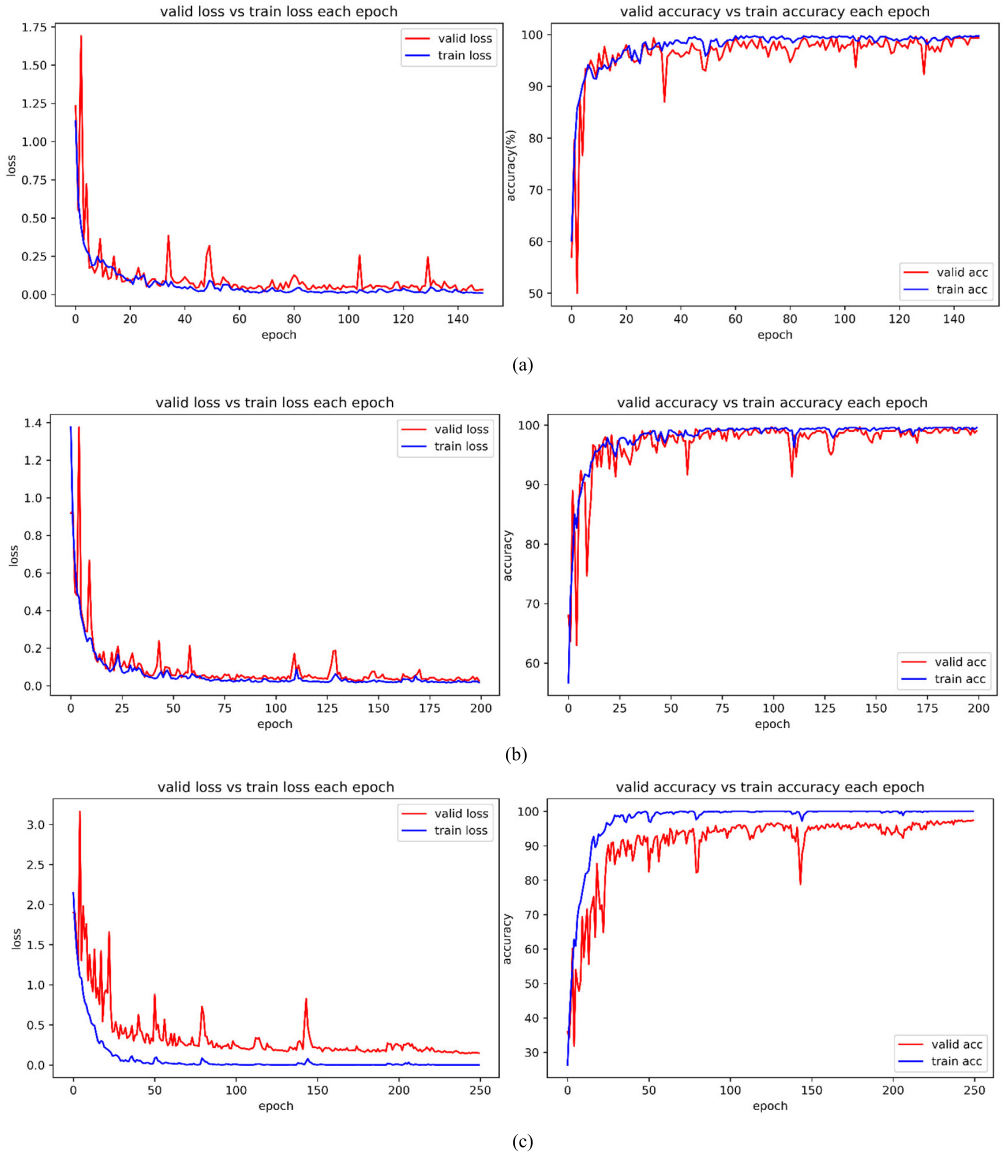


Fig. 14. Comparison of loss and accuracy in the search process. (a) Bayview Park. (b) Recology. (c) Houston.

TABLE V
COMPARISON OF ABLATION EXPERIMENTS ON THREE DATASETS

Dataset	Index	ACNAS	ACNAS-CPA	ACNAS-LS	ACNAS-CPA-LS
Bayview Park	OA(%)	96.31±0.49	96.65±0.82	96.94±0.76	97.32±0.57
	AA(%)	96.68±1.05	97.02±0.94	97.35±1.17	97.66±1.31
	K×100	95.79±0.85	96.19±0.69	96.52±0.91	96.87±0.69
Recology	OA(%)	96.97±0.67	97.37±0.84	97.65±0.62	98.08±0.76
	AA(%)	97.54±1.46	97.82±1.25	97.96±1.37	98.33±1.41
	K×100	96.73±0.64	97.04±0.54	97.17±0.82	97.51±0.58
Houston	OA(%)	96.04±0.47	96.39±0.18	96.65±0.14	97.12±0.21
	AA(%)	96.77±0.27	96.97±0.39	97.29±0.27	97.64±0.31
	K×100	95.83±0.13	96.31±0.22	96.46±0.39	96.85±0.28

class in the Bayview Park dataset, parking lot class and seawater class in the Recology dataset, and soil class and parking lot 1 class in the Houston dataset.

In order to show the classification results more clearly, we give the classification maps of seven methods on three LiDAR datasets. As shown in Figs. 10–12, we evaluate the impact of different methods on classification accuracy from a visual point of view. Obviously, compared with other models, ACNAS-CPA-LS obtains less noise scattering points in the class with large areas, which means that it can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF THE CONSUMING TIME OF CPA AND COSINE ANNEALING IN EACH STAGE

Dataset	Methods	Search Time(h)	Train Time(m)	Test Time(s)	Par.(M)
Bayview Park	ACNAS	2.9	8.5	8.6	0.094
	ACNAS-CPA	2.7	7.2	8.5	0.094
Recology	ACNAS	3.5	11.7	9.1	0.304
	ACNAS-CPA	3.2	10.5	9.1	0.304
Houston	ACNAS	4.9	14.8	10.2	0.508
	ACNAS-CPA	4.6	12.6	10.2	0.508

obtain higher classification accuracy in this class. In addition, we can also see that the automatically designed CNN has less misclassification area than the manually designed CNN model, which proves the feasibility of automatic architecture search CNN model. Compared with the ground-truth map, it can be seen that the proposed method can get more accurate classification results, which further proves the effectiveness of ACNAS-CPA-LS in the LiDAR data classification task.

E. Optimal Architecture Analysis

In our experiment, we use the DARTS search strategy and a series of measures such as the proposed search space to search the corresponding optimal network architecture on three LiDAR datasets. Fig. 13 shows the structure diagram of the cell finally searched on the three datasets. In the search phase, two kinds of cells can be searched: normal cell and reduction cell. In the normal cell, the step size of all operations is 1, so the size of the output characteristic graph remains unchanged. For the reduction cell, it is inserted at 1/3 and 2/3 of the network depth. In order to reduce the spatial resolution of the feature map, the step size of the operation of reducing the edge connected to the input node in the cell is set to 2.

From Fig. 13, we can find that the cells searched by this method retain many deeper connections (such as convolution operation and attention module) rather than shallow connections (such as skip_connect), so our model contains more learnable parameters and the expression ability is stronger. In addition, we can also see that the number of attention modules accounts for more than half of the number of operations between nodes. Therefore, our method has stronger ability to extract features, learn more effective features, and build more layers, so as to make the final model deeper and obtain better classification results.

Fig. 14 shows the loss and accuracy of the architecture corresponding to the three datasets in the search phase. With the increase of the number of epochs, the loss and accuracy will change accordingly. The loss in the training set and validation set is decreasing, and at the same time, the classification accuracy is improving. We can see that the corresponding architectures on the three datasets basically reach the convergence state when the epoch is about 60, and the optimal accuracy of the training set and validation set is still improving after the architecture converges. The main reason is that the parameters of the architecture inherited from the supernet are continuously optimized, and the supernet has been trained until the maximum number of epochs is reached.

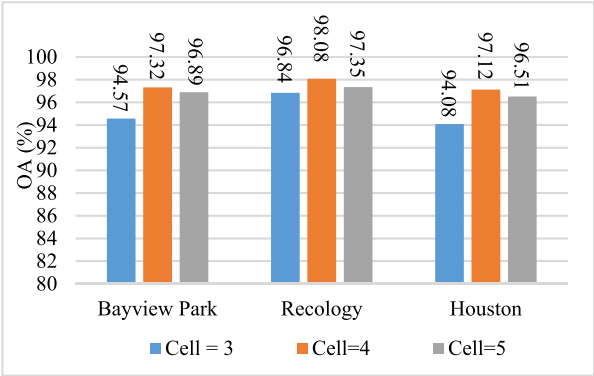


Fig. 15. Classification results of different numbers of cells.

Therefore, it is verified that the architecture in this article is convergent.

F. Ablation Study Analysis

A large number of ablation experiments were performed to verify the effectiveness of the proposed method, as shown in Table V. When we only use the ACNAS model to classify LiDAR data, its OA on the three datasets is only 96.31%, 96.97%, and 96.04%. When ACNAS is combined with CPA alone (ACNAS-CPA), the OA on the three datasets is increased by 0.34%, 0.42%, and 0.35%, which proves that CPA can slightly improve the classification performance of the model. When ACNAS is combined with label smoothing alone (ACNAS-LS), the OA on the three datasets is improved by 0.63%, 0.68%, and 0.61%, indicating that label smoothing is more effective than CPA in improving classification accuracy. Finally, when ACNAS is combined with these two strategies (ACNAS-CPA-LS), it achieves the highest classification accuracy on both datasets, which further proves the effectiveness of these two strategies in improving the classification performance of the model.

In order to verify that CPA can reduce the time consumed in various stages of the model, we compared it with the most commonly used cosine annealing in the NAS field. The experimental results are shown in Table VI. The experimental results show that CPA greatly reduces the time consumed by the model in the search stage and evaluation stage, and this strategy does not increase the number of parameters in the model, which proves its effectiveness in reducing the calculation cost.

In addition, we analyze the parameters of the number of cells (Cell = 3–5) and the number of nodes in each Cell

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

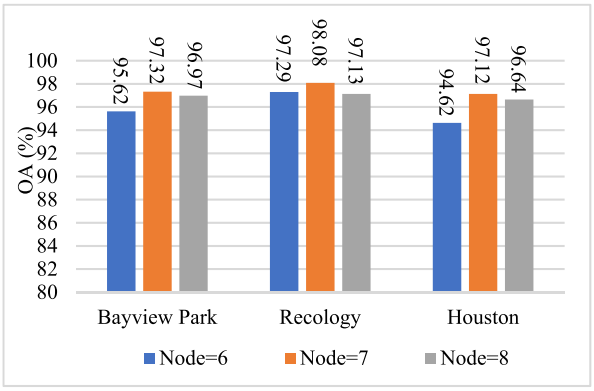


Fig. 16. Classification results of different numbers of intermediate nodes.

(Node = 6–8) used in the evaluation stage. As can be seen from Figs. 15 and 16, when the number of cells or intermediate nodes is small, the OA value obtained is low. However, when there are many cells or intermediate nodes, the final model does not necessarily lead to better performance. The main reason is that the model with a simple structure may not be able to extract the deep features, while the complex model may have the problem of fitting under the limited training samples. Thus, we set the number of cells to 3 and the number of nodes in the cell to 7.

V. Conclusion

In this article, we propose a neural network architecture search method for LiDAR DSM classification, called ACNAS-CPA-LS. It contains a convolution operation and attention module to form a new search space and then uses the gradient-based DARTS strategy to find the cell structure with attention module, so as to achieve an optimal model that can adapt to each kind of data. In addition, we propose a learning rate adjustment strategy, which can effectively reduce the time required to improve the classification performance of the model and further improve the efficiency of architecture search and the classification accuracy of the model. In addition, we also use an architecture regularization scheme to avoid overfitting, so as to enhance the robustness of the model. We conduct experiments on three public LiDAR datasets and compare ACNAS-CPA-LS with six methods to verify the effectiveness of the proposed method. The experimental results show that our method is more competitive. OA on Bayview Park, Recology, and Houston datasets reaches 97.32%, 98.08%, and 97.12%, respectively, which is better than two classical methods and four CNN-based models designed manually and automatically.

Our work opens a new way for LiDAR DSM classification tasks. In the future work, we will further develop new NAS algorithms to automatically design an efficient LiDAR data classification model and propose more effective search space to adapt to more complex remote sensing image classification tasks.

References

[1] S. Y. You, J. H. Hu, U. Neumann, and P. Fox, "Urban site modeling from LiDAR," in *Proc. Int. Conf. Comput. Sci. Appl. (ICCSA)*, Montreal, QC, Canada, 2003, May 18-21.

[2] V. I. Chernook, A. N. Vasilyev, Y. A. Goldin, B. A. Gureev, V. S. Goryainov, and A. A. Buznikov, "Oceanological monitoring of fishing areas using lidars," in *Proc. Int. Conf. Laser Opt.*, St. Petersburg, Russia, Jun. 2014, p. 1.

[3] Q. Ma, Y. Su, and Q. Guo, "Comparison of canopy cover estimations from airborne LiDAR, aerial imagery, and satellite imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 9, pp. 4225–4236, Sep. 2017, doi: 10.1109/JSTARS.2017.2711482.

[4] Y. Chen, L. Cheng, M. Li, J. Wang, L. Tong, and K. Yang, "Multiscale grid method for detection and reconstruction of building roofs from airborne LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 10, pp. 4081–4094, Oct. 2014, doi: 10.1109/JSTARS.2014.2306003.

[5] F. Scotti, D. Onori, M. Scaffardi, E. Lazzeri, A. Bogoni, and F. Laghezza, "Multi-frequency lidar/radar integrated system for robust and flexible Doppler measurements," *IEEE Photon. Technol. Lett.*, vol. 27, no. 21, pp. 2268–2271, Nov. 1, 2015, doi: 10.1109/LPT.2015.2461458.

[6] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "PointNet++: Deep hierarchical feature learning on point sets in a metric space," 2017, *arXiv:1706.02413*.

[7] L. Zhang and L. Zhang, "Deep learning-based classification and reconstruction of residential scenes from large-scale point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1887–1897, Apr. 2018, doi: 10.1109/TGRS.2017.2769120.

[8] T. Hackel, J. D. Wegner, and K. Schindler, "Fast semantic segmentation of 3D point clouds with strongly varying density," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3 pp. 177–184, Jul. 2016.

[9] Y. Song, J. Imanishi, H. Hashimoto, A. Morimura, Y. Morimoto, and K. Kitada, "Spectral correction for the effect of crown shape at the single-tree level: An approach using a lidar-derived digital surface model for broad-leaved canopy," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 2, pp. 755–764, Feb. 2013, doi: 10.1109/TGRS.2012.2204404.

[10] P. Ghamisi and B. Höfle, "LiDAR data classification using extinction profiles and a composite kernel support vector machine," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 659–663, May 2017, doi: 10.1109/LGRS.2017.2669304.

[11] L. Naidoo, M. A. Cho, R. Mathieu, and G. Asner, "Classification of savanna tree species, in the region, by integrating hyperspectral and LiDAR data in a random forest data mining environment," *ISPRS J. Photogramm. Remote Sens.*, vol. 69, pp. 167–179, Apr. 2012.

[12] A. Wang, X. He, P. Ghamisi, and Y. Chen, "LiDAR data classification using morphological profiles and convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 774–778, May 2018, doi: 10.1109/LGRS.2018.2810276.

[13] A. Wang, M. Wang, K. Jiang, L. Zhao, and Y. Iwahori, "A novel lidar data classification algorithm combined densenet with STN," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Yokohama, Japan, Jul. 2019, pp. 2483–2486.

[14] X. He, A. Wang, P. Ghamisi, G. Li, and Y. Chen, "LiDAR data classification using spatial transformation and CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 125–129, Jan. 2019, doi: 10.1109/LGRS.2018.2868378.

[15] A. Wang, M. Wang, K. Jiang, M. Cao, and Y. Iwahori, "A dual neural architecture combined SqueezeNet with OctConv for LiDAR data classification," *Sensors*, vol. 19, no. 22, p. 4927, Nov. 2019, doi: 10.3390/s19224927.

[16] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Ses.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017, doi: 10.1109/JSTARS.2016.2634863.

[17] H. Wu, M. Cao, A. Wang, and M. Wang, "Classification of LiDAR data combined octave convolution with capsule network," *IEEE Access*, vol. 8, pp. 16155–16165, 2020, doi: 10.1109/ACCESS.2020.2965278.

[18] Y. Bazi, M. M. Al Rahhal, H. Alhichri, and N. Alajlan, "Simple yet effective fine-tuning of deep CNNs using an auxiliary classification loss for remote sensing scene classification," *Remote Sens.*, vol. 11, no. 24, p. 2908, Dec. 2019, doi: 10.3390/rs11242908.

[19] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, and A. Rabinovich, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 27–30.

[21] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.

[22] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[23] A. Wang, Y. Li, K. Jiang, L. Zhao, and Y. Iwahori, "Lidar data classification algorithm based on generative adversarial network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Yokohama, Japan, Jul. 2019, pp. 2487–2490.
[24] A. Wang, D. Xue, H. Wu, and Y. Iwahori, "LiDAR data classification based on improved conditional generative adversarial networks," *IEEE Access*, vol. 8, pp. 209674–209686, 2020.
[25] L. Wang, X. Bai, and F. Zhou, "Few-shot SAR ATR based on conv-BiLSTM prototypical networks," in *Proc. 6th Asia–Pacific Conf. Synth. Aperture Radar (APSAR)*, Nov. 2019, pp. 1–5.
[26] L. Wang, X. Bai, C. Gong, and F. Zhou, "Hybrid inference network for few-shot SAR automatic target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9257–9269, Nov. 2021.
[27] M. Yang, X. Bai, L. Wang, and F. Zhou, "Mixed loss graph attention network for few-shot SAR target classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5216613.
[28] T. Elsken, J. H. Metzen, and F. Hutter, "Neural architecture search: A survey," *J. Mach. Learn. Res.*, vol. 20, no. 55, pp. 1–21, 2019.
[29] B. Baker, O. Gupta, N. Naik, and R. Raskar, "Designing neural network architectures using reinforcement learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017. [Online]. Available: https://arxiv.org/abs/1611.02167
[30] B. Zoph and Q. V. Le, "Neural architecture search with reinforcement learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017. [Online]. Available: https://arxiv.org/abs/1611.01578
[31] H. Pham, M. Y. Guan, B. Zoph, Q. V. Le, and J. Dean, "Efficient neural architecture search via parameter sharing," in *Proc. Int. Conf. Mach. Learn.*, 2018. [Online]. Available: https://arxiv.org/abs/1802.03268
[32] S. Xie, H. Zheng, C. Liu, and L. Lin, "SNAS: Stochastic neural architecture search," 2018, *arXiv:1812.09926*.
[33] T. Veniat, O. Schwander, and L. Denoyer, "Stochastic adaptive neural architecture search for keyword spotting," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Brighton, U.K., May 2019, pp. 2842–2846.
[34] T. Domhan, J. T. Springenberg, and F. Hutter, "Speeding up automatic hyperparameter optimization of deep neural networks by extrapolation of learning curves," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, Jul. 2015, pp. 3460–3468.
[35] H. Mendoza, A. Klein, M. Feurer, J. T. Springenberg, and F. Hutter, "Towards automatically-tuned deep neural networks," in *Automated Machine Learning*. Cham, Switzerland: Springer, 2019, pp. 135–149.
[36] H. Pham, M. Guan, B. Zoph, Q. V. Le, and J. Dean, "Efficient neural architecture search via parameters sharing," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Stockholm, Sweden, Jul. 2018, pp. 6522–6531.
[37] L. Xie and A. Yuille, "Genetic CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 1388–1397.
[38] I. Bello, B. Zoph, V. Vasudevan, and Q. V. Le, "Neural optimizer search with reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, vol. 70, 2017, pp. 459–468.
[39] Z. Zhong, J. Yan, W. Wu, J. Shao, and C.-L. Liu, "Practical block-wise neural network architecture generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2423–2432.
[40] H. Liu, K. Simonyan, and Y. Yang, "DARTS: Differentiable architecture search," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019. [Online]. Available: https://arxiv.org/abs/1806.09055
[41] H. Cai, L. Zhu, and S. Han, "ProxylessNAS: Direct neural architecture search on target task and hardware," in *Proc. Int. Conf. Learn. Represent.*, New Orleans, LA, USA, 2019. [Online]. Available: https://arxiv.org/abs/1812.00332
[42] E. Real, A. Aggarwal, Y. Huang, and Q. V. Le, "Regularized evolution for image classifier architecture search," in *Proc. 33rd AAAI Conf. Artif. Intell. (AAAI)*, Palo Alto, CA, USA, 2019, pp. 4780–4789.
[43] B. Zoph and Q. V. Le, "Neural architecture search with reinforcement learning," 2016, *arXiv:1611.01578*.
[44] H. Cai, T. Chen, W. Zhang, Y. Yu, and J. Wang, "Efficient architecture search by network transformation," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 1–8.
[45] H. Xie and Y. Chen, "LiDAR data classification based on automatic designed CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 9, pp. 1665–1669, Sep. 2021.
[46] F. Wang *et al.*, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6450–6458.
[47] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.
[48] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141, doi: 10.1109/CVPR.2018.00745.
[49] J. Park, S. Woo, J. Y. Lee, and I. S. Kweon, "BAM: Bottleneck attention module," in *Proc. BMVC*, 2018. [Online]. Available: https://arxiv.org/abs/1807.06514
[50] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *ECCV*, 2018, pp. 3–19.
[51] D. Misra, T. Nalamada, A. U. Arasanipalai, and Q. Hou, "Rotate to attend: Convolutional triplet attention module," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3138–3147.
[52] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.
[53] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.
[54] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, San Juan, Pueto Rico, 2016. [Online]. Available: https://arxiv.org/abs/1511.07122



Aili Wang (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in information and signal processing from the Harbin Institute of Technology, Harbin, China, in 2002, 2004, and 2008, respectively.

She has been a Visiting Professor in the research of 3-D polyp reconstruction with the Computer Science Laboratory, Chubu University, Kasugai, Japan, in 2014. She is the author of two books and more than 80 articles. Her research interests include deep learning, space intelligent remote sensing, and information processing.



Dong Xue received the B.S. degree in communication engineering from the Harbin University of Science and Technology, Harbin, China, in 2019, where he is currently pursuing the master's degree in communication and information system.

He has published two articles in the IEEE ACCESS and *Chinese Journal of Liquid Crystals and Displays*. His research interests include light detection and ranging (LiDAR) data classification, generative adversarial networks, and neural architecture search.



Haibin Wu received the B.S. and M.S. degrees from the Harbin Institute of Technology, Harbin, China, in 2000 and 2002, respectively, and the Ph.D. degree in measuring and testing technologies and instruments from the Harbin University of Science and Technology, Harbin, in 2008.

From 2014 to 2015, he was a Visiting Scholar with the Robot Perception and Action Laboratory, University of South Florida, Tampa, FL, USA. He is the author of three books, more than 40 articles, and more than 20 inventions. His research interests include robotic vision, visual measuring and image processing, medical virtual reality, and photoelectric testing.



Yanfeng Gu (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2005.

He joined the School of Electronics and Information Engineering, HIT, as a Lecturer, where he was appointed as an Associate Professor in 2006. Meanwhile, he was enrolled in first Outstanding Young Teacher Training Program with HIT. From 2011 to 2012, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of California at Berkeley, Berkeley, CA, USA. He is currently a Professor with the Department of Information Engineering, HIT. He has authored or coauthored more than 100 peer-reviewed articles and four book chapters. He is the inventor or coinventor of 20 patents. His research interests include space intelligent remote sensing and information processing, multimodal hyperspectral remote sensing, and spaceborne time series image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:38:12 UTC from IEEE Xplore. Restrictions apply.